

PEDRO H. R. DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2023 – present 📍 Rio de Janeiro, Brazil

Technical Degree in Informatics

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 February. 2020 – December. 2023 📍 Rio de Janeiro, Brazil

RESEARCH PROJECTS

Understanding Dark Matter from ExtraGalactic Shocks

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 September. 2022 – March. 2023 📍 Rio de Janeiro, Brazil

Junior Scientific Initiation Scholarship - CNPq
Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2023 – September. 2023 📍 Rio de Janeiro, Brazil

Volunteer in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2023 – February. 2024 📍 Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - IFF
Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning 2025

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 October. 2025 – present 📍 Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2020 📍 Rio de Janeiro, Brazil

Operational Systems

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2021 📍 Rio de Janeiro, Brazil

Server Administration

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
📅 August. 2022 📍 Rio de Janeiro, Brazil

RESEARCH INTERESTS

- Stars
- Milky Way
- Galactic Archaeology
- Star Evolution
- Quantum Mechanics
- Computational Physics
- ML
- AI
- Physics
- Astrophysics Computational
- Deduction of Equations
- Animations
- Simulations

EVENTS

Science Fairs

21st Brazilian Science and Engineering Fair

Poster [10]
USP, São Paulo

March. 2023 São Paulo, Brazil

40th International Science and Technology Fair

Poster [12]
Fundação Librelato, Novo Hamburgo

October. 2023 Rio Grande do Sul, Brazil

IX Brazilian Science Initiation Fair

Poster [9]
Pomerode

September. 2024 Santa Catarina, Brazil

XVI Scientific and Technological Exhibition of the Açaí Institute

Poster [7]
UFPA, Belém

December. 2025 Pará, Brazil

Conferences

44° CONUBES

Arena Nilson Nelson, Brasília
May. 2022 Distrito Federal, Brazil

XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies

Oral Presentation [8]
UENF, Campos dos Goytacazes
May. 2024 Rio de Janeiro, Brazil

XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies

Oral Presentation [6]
UENF, Campos dos Goytacazes
May. 2025 Rio de Janeiro, Brazil

XLVIII Annual Meeting of the Brazilian Astronomical Society

Poster [1, 14]
Hotel Glória, Caxambu
September. 2025 Minas Gerais, Brazil

XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies

Talk [3]
UENF, Campos dos Goytacazes
May. 2026 Rio de Janeiro, Brazil

78ª Annual Meeting of Brazilian Society to Science Progress

Poster [4]
UFF Campus Gragoatá, Niterói
July. 2026 Rio de Janeiro, Brazil

Winter School of Astrophysics

Poster [5]

Observatório Nacional, Rio de Janeiro

July. 2026 Rio de Janeiro, Brazil

XLIX Annual Meeting of the Brazilian Astronomical Society

Poster [2]
Natal

October. 2026 Rio Grande do Norte, Brazil

Teaching, Research and Extension Congress (CONEPE)

Presentation
IFF Campus Guarus, Campos dos Goytacazes

September. 2026 Rio de Janeiro, Brazil

Academic Weeks

VIII Knowledge Showcase and I Job Fair

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 December. 2020 📍 Rio de Janeiro, Brazil

IX Knowledge Showcase and II Job Fair

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 October. 2021 📍 Rio de Janeiro, Brazil

I Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 May. 2021 📍 Rio de Janeiro, Brazil

X Knowledge Showcase and III Job Fair

Talk
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 October. 2022 📍 Rio de Janeiro, Brazil

II Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 May. 2022 📍 Rio de Janeiro, Brazil

XI Knowledge Showcase and IV Job Fair

Poster [[11](#), [13](#)]
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 September. 2023 📍 Rio de Janeiro, Brazil

III Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 August. 2023 📍 Rio de Janeiro, Brazil

XII Knowledge Showcase and V Job Fair

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 October. 2024 📍 Rio de Janeiro, Brazil

IV Week of Computer Engineering and Management

Talk & Organizer
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

📅 October. 2024 📍 Rio de Janeiro, Brazil

PUBLICATIONS

 [Link Nasa ADS: NASA ADS - Pedro H. R. de Andrade](#)

Proceedings

- [1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.
- [2] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE ». in: *Boletim da Sociedade Astronômica Brasileira*. Natal, RN, 2026.
- [3] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4 ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.
- [4] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Anais da 78ª Reunião Anual da Sociedade Brasileira para o Progresso da Ciência (SBPC)*. Niterói, RJ: UFF, 2026.
- [5] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Resumos da Escola de Inverno de Astrofísica do Observatório Nacional*. Rio de Janeiro, RJ: Observatório Nacional, 2026.
- [6] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecília Soja. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.
- [7] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.
- [8] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.
- [9] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.
- [10] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Análise de simulações dinâmicas de aglomerados de galáxias em fusão ». In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.
- [11] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 17.
- [12] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Fabrício Barros Gonçalves. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da MOSTRATEC 2023*. 2023, p. 102.
- [13] Gabriel Coelho de Souza Poubel et al. « MobFog no IFFMaker ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

- [14] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *arXiv e-prints* (Dec. 2025). DOI: [10.48550/arXiv.2512.10037](#).

AWARDS



IX FEBIC

3rd Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"



XVI MCTIA

1st Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

STATISTICS



3 Oral Presentations

National Congresses



7 Posters

National and International Events



9 Science Fairs/Congresses

National and International Events



10 Academic Weeks

Regional Events



PROGRAMMING



LANGUAGES

Portuguese



English



Francês



Spanish

